Dinah J. Hartmann

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Marine biologist and ecological statistician with background and/or training in biology, conservation, computation, bioacoustics and ecological statistics. PhD student researching harbour porpoise density using passive acoustic monitoring, animal tag and visual data, computational tools and ecological statistics.

Education

University of St Andrews, St Andrews, Scotland **2024-2028**

PhD in Statistics

Project: Estimating spatial and temporal patterns in density of harbour porpoises in the North Sea using datasets from passive acoustic ocean observation networks with data flow to EU Digital Twin Ocean database.

Lund University, Lund, Sweden **2022-2024**

Master of Science in Biology, Conservation Biology, 120 Credits, VG (distinction)

Relevant courses such as “Global Ecosystem Dynamics”, “Population and Community Ecology”, “Conservation Biology” and “Biological Monitoring”
60-credit MSc thesis on the density of harbour porpoises (*Phocoena phocoena*) using drones

University College Dublin, Dublin, Ireland **2018-2022**

Bachelor of Science in Environmental Biology, 240 Credits, GPA 3.81 (First-class honours)

Relevant courses such as “Ornithology”, “Animal Behaviour”, “Conservation of Biodiversity”, “Working with Biological Data”, “Population Genetics”, “Wildlife Conservation and Fisheries Management”, “Diversity of Vertebrates”, “Evolutionary Biology” and “Wildlife Habitat Modelling”

Research

Center for Research into Ecological and Environmental Modelling, University of St Andrews, St Andrews, Scotland **2024-2028**

PhD thesis advised by Len Thomas, Peter Tyack and Jamie Macaulay: Processing and analysing acoustic and movement tag data in PAMGuard and MATLAB to determine the probability of detecting harbour porpoise echolocation clicks on click detectors.

Department of Biology, Lund University, Lund, Sweden **2023-2024**

Master of Science thesis advised by Per Carlsson and Johanna Stedt: Extensive field-work and drone-piloting experience, including focal follows of animals, programming drone paths and analysing the data in, e.g., DroneVideoMeasure, R and Excel.

Research assistance position: Supervising students in the field and independently collecting drone focal follow data of harbour porpoises.

School of Biology and Environmental Science, University College Dublin, Dublin, Ireland **2022**

Bachelor of Science thesis advised by Tancredi Caruso on the effects of drought on soil biodiversity: Ecological laboratory work to determine soil biodiversity, extracting and preserving biota, analysing them under microscopes.

Teaching and Outreach

School of Mathematics and Statistics, University of St Andrews, St Andrews, Scotland **2025-2028**

MT2508 Statistical Inference: Lead demonstrator for statistical programming in R, provided R statistics and programming support to students **2025**

MT2508 Statistical Inference: Tutor in statistics, Prepared and held small group tutorials, marked assignments and gave feedback **2025**

Department of Biology, Lund University, Lund, Sweden **2023-2024**

Teaching drone methodology (piloting, safety, marine mammal focal follows and flight path programming to Master's students

TV-documentary segment on harbour porpoise drone research on ARTE ("Night Trains – Von Berlin nach Stockholm")

Feature in 'Sveriges Natur', the biggest Swedish nature magazine ("Kampen för att rädda tumlaren")

Smaller newspaper articles in 'tidningensyre' ("Snärjs i fisknät – kan tumlarna räddas från dödsdansen?" and 'Helsingborg dagblad' ("Barndomens kärlek till Sveriges enda val blev hennes karriär")

Seal Conservancy Norden-Norddeich, Norden, Germany **2021**

Voluntary work in environmental education and outreach: Insights into seal conservation and monitoring, public outreach talks, teaching museum visitors in seal conservancy work via small talks and personal interactions

Private Tutoring in Mathematics and French, Essen, Germany **2016-2018**

Prepared students for A-level Mathematics and French in independently organised lessons

Secondary school Maria-Wächtler-Gymnasium, Essen, Germany **2016-2017**

Tutoring programme for children from refugee areas

Secondary school Maria-Wächtler-Gymnasium, Essen, Germany **2016**

Internship as teacher

Publications

Hartmann, D., Palmqvist, V. and Stedt, J. (2026), Programmed unmanned aerial vehicles show great potential for monitoring marine megafauna in specific areas of interest. Remote Sens Ecol Conserv. https://doi.org/10.1002/rse2.70043

Talks and Posters

Hartmann, D., Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L. (January 2026). The probability of detecting harbour porpoise echolocation clicks. [Invited Talk]. Aarhus University Ecoscience Group, Roskilde, Denmark.

Hartmann, D., Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L. (November 2025). The probability of detecting harbour porpoise echolocation clicks. [Invited talk]. Marine Alliance for Science and Technology Scotland, Glasgow, Scotland.

Hartmann, D., Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L. (October 2025). The probability of detecting harbour porpoise echolocation clicks. [Invited talk]. Scottish Oceans Institute, St Adnrews, Scotland.

Hartmann, D., Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L. (September 2025). The probability of detecting harbour porpoise echolocation clicks. [Poster]. International Bioacoustics Congress, Kerteminde, Denmark.

Hartmann, D., Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L. (September 2025). The probability of detecting harbour porpoise echolocation clicks. [Invited Talk]. Aarhus University Bioacoustics Group, Aarhus, Denmark.

Hartmann, D., Palmqvist, V. & Stedt, J. (May 2025). En route to standardisation: Porpoise monitoring using a programmed UAV. [Poster]. European Cetacean Society 36th conference, Ponta Delgada, Azores, Portugal.

Hartmann, D., Macaulay, J. D. J., Tyack, P., & Thomas, L. (January 2025). Harbour porpoise click detection. [Poster]. UK and Ireland regional student chapter for marine mammology student conference, St Andrews, Scotland.

Hartmann, D., Stedt, J., & Carlsson, P. (March 2024). Spatio-temporal distribution of harbour porpoises around Kullaberg nature reserve. [Invited Talk]. Danish Marine Mammal Society Symposium, Denmark.

Awards, Grants, Scholarships

Doctoral studentship, EU funded as part of the DTO-Bioflow project (~120,000 gbp) **2024-2028**

Environmental Biology Medal for highest overall Bachelor's GPA **2022**

Environmental Biology Stage 4 Scholarship for highest class GPA (1,000 euro) **2021**

Skills and Trainings

Scottish Mathematical Sciences Training Centre **2024-2025**

Modern Regression and Simulation methods and Bayesian statistics

University of Oxford, Oxford, England **December 2024**

Academy for PhD training in statistics (Statistical Inference and computing)

University of New Hampshire, Durham, USA **June 2024**

SeaBASS summer school in bioacoustics

Computing, quantitative

R

RStan

MATLAB

ArcGIS Pro

QGIS

PAMGuard

Raven

SPSS

Other research skills

A1-A3 Drone License, drone piloting and programming

LaTeX/Overleaf

Microsoft office applications

German (native proficiency)

French

Spanish

Laboratory skills (soil drying, extraction of biota, microscopes, DNA extraction)